Lecture 3

Saturday, 22 August 2026 11:23 AM

Prob. Space $(\Omega, \mathcal{F}, \mathbb{P})$

$\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if A, B disjoint

(1) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

(2) $A \subseteq B$, $\mathbb{P}(A) \leq \mathbb{P}(B)$

(3) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Inclusion - Exclusion

$$\mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots (-1)^{n+1} \cdot \mathbb{P}(A_1 \cap A_2 \dots \cap A_n)$$

($\binom{n}{2}$ terms in the $i<j$ sum)

100 balls, 20 bins. $\mathbb{P}(\exists \text{ an empty bin}) = \mathbb{P}\left(\bigcup_{i=1}^{20} A_i\right)$

Bonferroni Ineq

→ Union Bound - $\mathbb{P}(\cup A_i) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)$

→ 2nd Bon. ineq - $\mathbb{P}(\cup A_i) \geq \sum \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j)$

3rd Bon. ineq - $\mathbb{P}(\cup A_i) \leq \sum \mathbb{P}(A_i) - \sum \mathbb{P}(A_i \cap A_j) + \sum \mathbb{P}(A_i \cap A_j \cap A_k)$

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100 balls, 20 bins

$$P\left(\bigcup_{i=1}^{20} A_i\right) \leq \sum_{i=1}^{20} P(A_i)$$

$$P(A_1) = \left(\frac{19}{20}\right)^{100}$$

$P(A_2)$

$$= 20 \times \left(\frac{19}{20}\right)^{100} \approx .118$$

$$P(\cup A_i) \geq \sum P(A_i) - \sum_{i<j} P(A_i \cap A_j)$$

$$.118 - \binom{20}{2} \times \left(\frac{18}{20}\right)^{100} \approx .1133$$

$$P(A_1 \cap A_2) = \left(\frac{18}{20}\right)^{100}$$

$$.1133 \leq P(\cup A_i) \leq .118$$

$P(\cup A_i) = .1134$

Discrete Prob. space

Ω is finite or Countable

$\Omega = \{1, 2, 3 \ldots\}$

$i \in \Omega$, $P(\{i\}) = p_i$

$P(S) = \sum_{i \in S} p_i$

$\Omega = \{1, 2, 3 \ldots\}$

$p_i = a_i$ $\quad \sum_{i=1}^{\infty} a_i = 1$

$P(\{1,2\}) = p_1 + p_2$

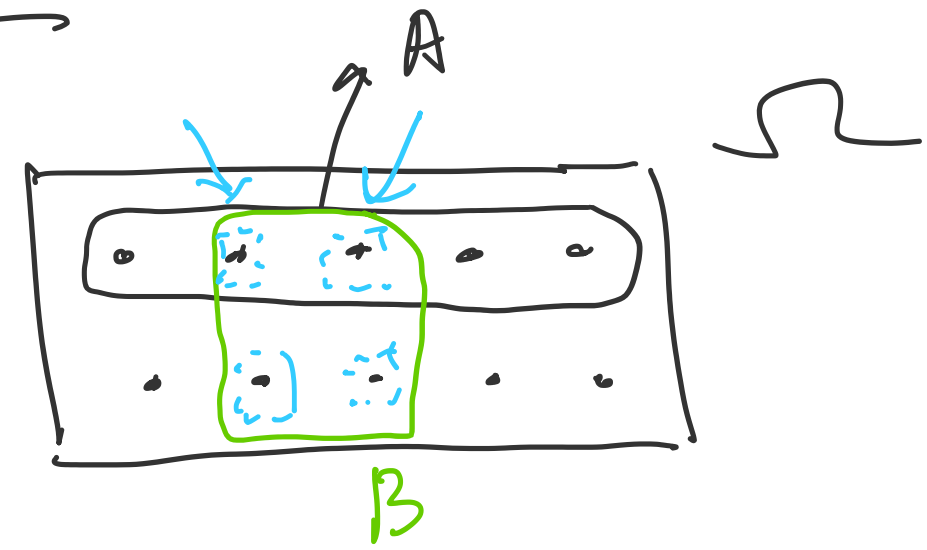
$P(\{3,4,5\}) = p_3 + p_4 + p_5$

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Conditional probability

Ω = 10 pts

$P(E) = \frac{|E|}{10}$, $E \subseteq \Omega$



$P(A) = \frac{5}{10} = \frac{1}{2}$, $P(B) = \frac{4}{10} = \frac{2}{5}$

Given B occurred, how likely is A to occur?

$$P(A|B) \triangleq \frac{N(B \cap A)/|\Omega|}{N(B)/|\Omega|} = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

Eg 2 coins C_1, C_2

C_1 → Fair

C_2 → Heads

Exp → Choose a coin, Toss it once.

$\Omega = \{(C_1, H), (C_1, T), (C_2, H), (C_2, T)\}$

P → $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{2}$ 0

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Events → $\mathcal{E}$ → observe Tails → $\{(C_1,T),(C_2,T)\}$

A_1 → Choose Coin 1 → $\{(C_1,T),(C_1,H)\}$

→ A_2 → Choose coin 2 → $\{(C_2,T),(C_2,H)\}$

$$P(\mathcal{E}) = \underbrace{P((C_1,T))}_{1/4} + \underbrace{P((C_2,T))}_{0} = \frac{1}{4}$$

$$P(A_1) = \frac{1}{2} = P(A_2)$$

P(observing Tails given chose coin 1)

$$P(\mathcal{E}|A_1) = \frac{P(\mathcal{E}\cap A_1)}{P(A_1)} = \frac{P(\{C_1,T\})}{1/2} = \frac{1/4}{1/2} = \frac{1}{2}$$

$$P(\mathcal{E}|A_2) = 0$$

$$= \frac{P(\mathcal{E}\cap A_2)}{P(A_2)} = \frac{P(\{C_2,TT\})}{P(A_2)} = 0$$

Properties of Conditional Prob.

1. Multiplicative Rule
2. Law of Total Prob.
3. Bayes Rule

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① Multiplicative Rule

$$P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$$

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1) \cdot \prod_{i=2}^{n} P(A_i \mid A_1 \cap A_2 \ldots \cap A_{i-1})$$

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② Law of Total Prob.

$$P(A) = \sum_{i \in I} P(A \cap B_i)$$

$$= \sum_{i \in I} P(B_i) \cdot P(A|B_i)$$



$\{B_i\}$ form a partition of Ω

$\bigcup B_i = \Omega$

B_i's disjoint, $B_i \cap B_j = \emptyset$

$I = \{1, 2, 3\}$

Eg C_1, C_2 $\mathcal{E} \rightarrow$ observe Tails.

$P(\mathcal{E})$ A_1, A_2 partition Ω

$\{(C_1, T), (C_1, H)\}$ $\{(C_2, H), (C_2, T)\}$

$$P(\mathcal{E}) = P(A_1) \cdot P(\mathcal{E}|A_1) + P(A_2) \cdot P(\mathcal{E}|A_2)$$

$$= \frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{2} \quad 0$$

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$= 1/4$

Toss coin 3 times

A, B, C
↓ ↓ ↘
Toss 1, Toss 2, Toss 3

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

Bayes Rule — Prior Belief
Data / Event
Update Belief

Given → A, $\{B_i\}_{i \in I}$ partition Ω

→ $P(B_i)$ → initial belief / prior

→ $P(B_i|A)$ → updated belief

Likelihood

$$P(B_i|A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(B_i) \cdot P(A|B_i)}{\sum_{j \in I} P(B_j) \cdot P(A|B_j)}$$

Eg → C_1, C_2, A_1, A_2, ε^c → observe Heads

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$$P(\mathcal{E}^c) = 1 - P(\mathcal{E}) = 1 - \frac{1}{4} = \frac{3}{4}$$

$P(A_1) = P(A_2) = \frac{1}{2}$

$\mathcal{E}^c$ is true

$$P(A_1 \mid \mathcal{E}^c) = \frac{P(A_1 \cap \mathcal{E}^c)}{P(\mathcal{E}^c)} = \frac{P(\{(T,H)\})}{3/4} = \frac{1/4}{3/4} = \frac{1}{3}$$

$$P(A_2 \mid \mathcal{E}^c) = \frac{P(A_2 \cap \mathcal{E}^c)}{P(\mathcal{E}^c)} = \frac{P(\{(T,H)\})}{3/4} = \frac{1/2}{3/4} = \frac{2}{3}$$

$(\Omega, \mathcal{F}, P)$ $B \in \mathcal{F}$ is some event $P(B) > 0$

$P_B(A) = P(A \mid B)$ for any event $A \in \mathcal{F}$

$P_B(\cdot)$ is also a valid prob measure

$P_B(\emptyset) = 0$, $P_B(\Omega) = 1$.

A_1, A_2 disj. $P_B(A_1 \cup A_2) = P_B(A_1) + P_B(A_2)$

Choose one Coin, Toss it twice Heads both times

$P(A_1 \mid \{H,H\})$

$P(A_2 \mid \{H,H\})$

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